

Seminar presentation – Pontifícia Universidade Católica do Paraná
20 August 2026

Shape and insulation domain optimization regarding wall energy efficiency

Who I am ?

I am William DUMONTAUD and I have been a  PhD student since 1st November 2024 at LaSIE.

- Supervisor :
Julien BERGER, CNRS researcher for LaSIE, UMR 7356.
- Co-Supervisor :
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Funding: ANR JCJC 2023 TOPS (Optimisation topologique et de forme des parois opaques de bâtiments) project.



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 - Governing equation
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My research topic

Shape and insulation domain optimization regarding wall energy efficiency.

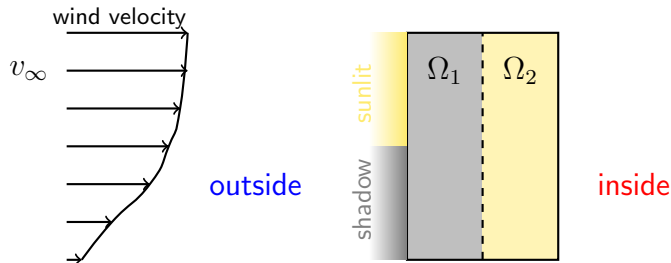


Figure 1: Illustration of non-uniform boundary conditions¹².

¹Rocha, Oliveira, and Mendes, "Experimental validation and comparison of direct solar shading calculations within building energy simulation tools", 2017.

²William H. Mcadams, *Heat Transmission*, 1954.

Global strategy : stage 1

Shape and **insulation domain optimization** regarding wall energy efficiency.

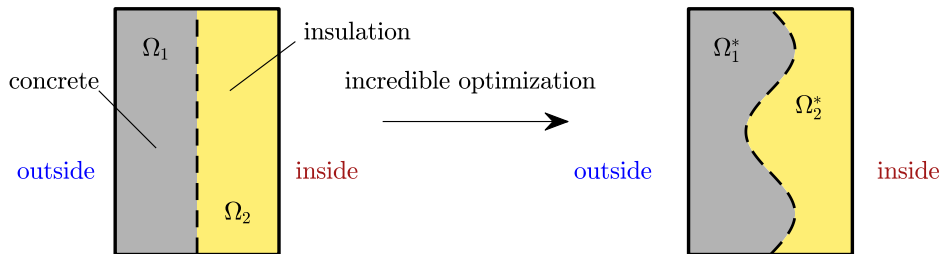


Figure 2: Wall domain insulation optimization³.

³Karashbayeva, Gasparin, and Berger, "Optimisation of the insulation domain for heat transfer in building wall", 2024.

Global strategy : stage 2

Shape and insulation domain **optimization** regarding wall energy efficiency.

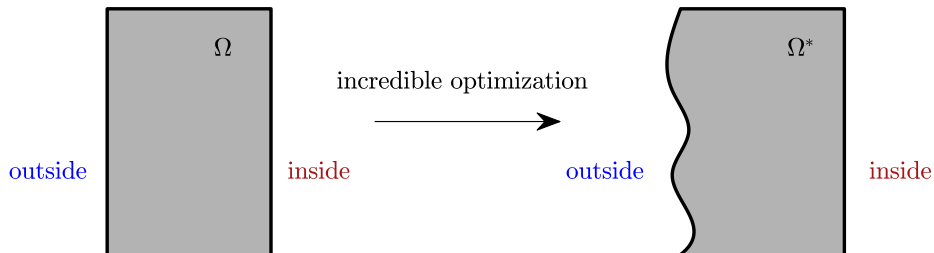


Figure 3: Wall shape optimization⁴.

⁴Alpar et al., "Shape optimization of the energy efficiency of building retrofitted facade", 2024.

Global strategy : stage 3

Shape and insulation domain optimization regarding wall energy efficiency.

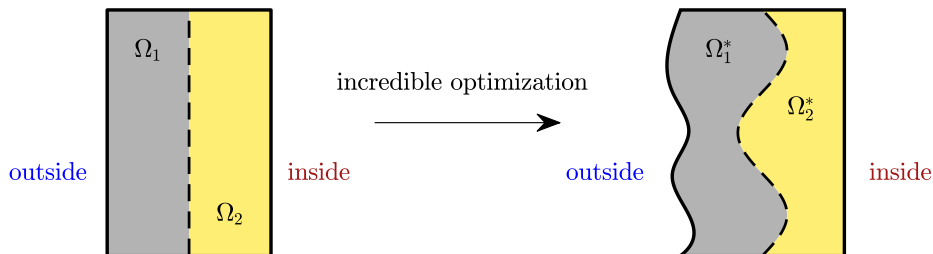


Figure 4: Wall shape & domain insulation optimization.

Global strategy : How to assess the *wall energy efficiency* ?

Shape and insulation domain optimization regarding **wall energy efficiency**.

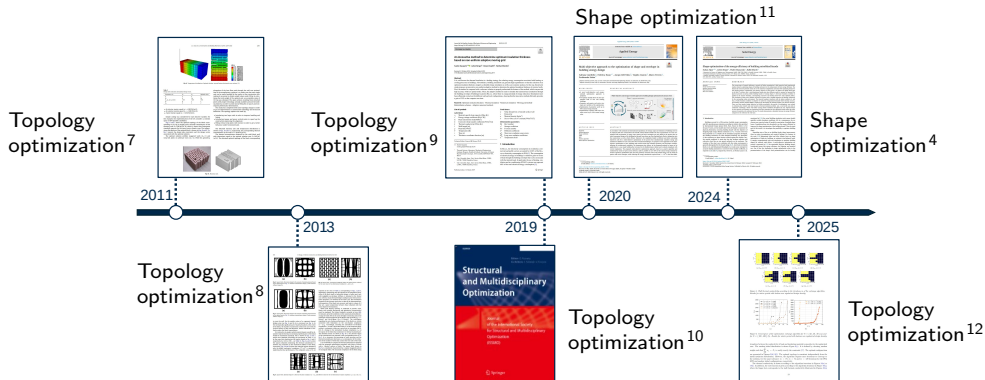
What physical quantities should be taken into account ?

- heat flux through the wall⁴³ ;
- surface temperatures ;
- thermodynamic approaches : *net consumed work* efficiency⁵, exergy destruction rate⁶;
- other quantities ...

⁵Strub et al., "Second law analysis of periodic heat conduction through a wall", 2005.

⁶Berger et al., "Thermodynamic analysis of the effect of mass transfer on a real building wall efficiency under climatic transient conditions", 2024.

State of the art



⁷Sousa et al., "Topology optimisation of masonry units from the thermal point of view using a genetic algorithm", 2011.

⁸Bruggi and Taliercio, "Design of masonry blocks with enhanced thermomechanical performances by topology optimization", 2013.

⁹Gasparin et al., "An innovative method to determine optimum insulation thickness based on non-uniform adaptive moving grid", 2019.

¹⁰Vantyghem et al., "Multi-material topology optimization involving simultaneous structural and thermal analyses", 2019.

¹¹Ciardiello et al., "Multi-objective approach to the optimization of shape and envelope in building energy design", 2020.

¹²Karashbayeva, Gasparin, and Berger, "Optimization of the insulation domain for heat transfer in building walls", 2025.

State of the art

Limitations – None of these approaches combines:

- two-dimensional approximation;
- transient regime solving;
- time and spatial variations of the boundary conditions;
- Taking into account the continuity of heat flux at the interface between two materials;

Manufacturing advances – New technical possibilities reinforce the potential of such an approach¹³.

¹³Gosselin et al., "Large-scale 3D printing of ultra-high performance concrete – a new processing route for architects and builders", 2016.

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Physical problem

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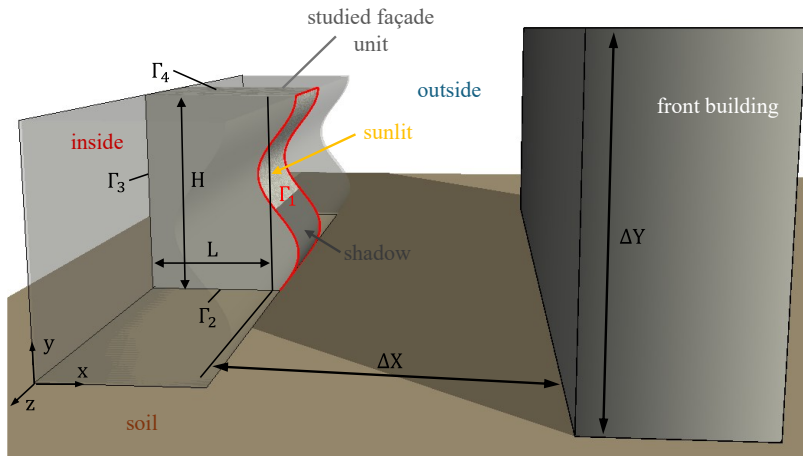


Figure 5: 3D illustration of the street canyon configuration showing the physical model.

Governing equations

2D thermal diffusion equation for transient conditions:

$$\frac{\partial}{\partial x} \left(\lambda(\mathbf{x}) \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(\lambda(\mathbf{x}) \frac{\partial T}{\partial y} \right) = c \frac{\partial T}{\partial t}, \quad \forall \mathbf{x} \in \Omega \times [0, t_f],$$

At the initial time, the domain is assumed to be in a state that satisfies the boundary conditions:

$$T = T_0(\mathbf{x}), \quad \forall \mathbf{x} \in \Omega, \quad t = 0.$$

Robin boundary conditions with space-time dependencies are taken into account:

$$-\lambda \nabla T \cdot \mathbf{n} = h_{\infty, j}(y, t) \cdot (T_{\infty, j}(t) - T(\mathbf{x}, t)) + q_{\infty, j}(\mathbf{x}, t), \quad \forall \mathbf{x} \in \Gamma_j, \quad j \in \{1, 2, 3, 4\}.$$

Equality of temperatures and continuity of the heat flux at the interface between two materials:

$$T|_{\mathbf{x}+\epsilon} = T|_{\mathbf{x}-\epsilon}, \quad \lambda(\mathbf{x}) \frac{\partial T}{\partial \mathbf{n}^+} \bigg|_{\mathbf{x}+\epsilon} = \lambda(\mathbf{x}) \frac{\partial T}{\partial \mathbf{n}^-} \bigg|_{\mathbf{x}-\epsilon}, \quad \forall \mathbf{x} \in \Gamma_i.$$

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External boundary conditions

Convective coefficient based on empirical correlation²:

$$h_{\infty,1}(y,t) = h_{10} + h_{11} \cdot \frac{v_{\infty}(t)}{v_0} \cdot \left(\frac{y}{y_0} \right)^{\beta},$$

Radiative heat flux:

$$q_{\infty,1}(\mathbf{x},t) = q^{sw}(\mathbf{x},t) + q^{\ell w}(\mathbf{x},t).$$

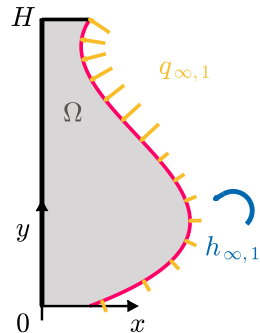


Figure 6: External boundary conditions.

External boundary conditions

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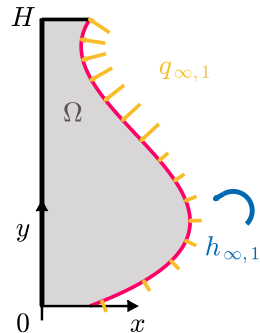


Figure 6: External boundary conditions.

External boundary conditions – short-wave modelling

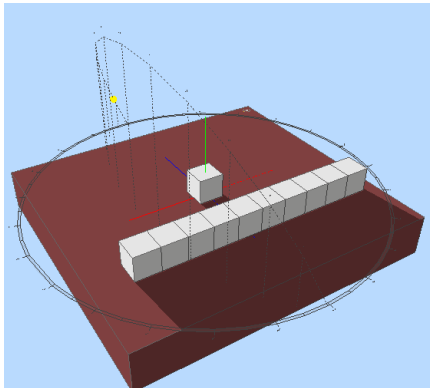


Figure 7: Modelling the impact of front building on short-waves using the pixel counting technique¹ in Domus.

Short-wave component:

$$q^{sw}(\mathbf{x}, t) = \alpha \left(\underbrace{q^{\text{dr}}(\mathbf{x}, t)}_{\text{direct}} + \underbrace{q^{\text{df}}(\mathbf{x}, t)}_{\text{diffuse}} + \underbrace{q^{\text{rf}}(\mathbf{x}, t)}_{\text{reflected}} \right),$$

Long-wave component:

$$\begin{aligned} q^{\ell w}(\mathbf{x}, t) = & \varepsilon \sigma T^4(\mathbf{x}, t) - \varepsilon \sigma \left(\underbrace{F_{1-g}(\mathbf{x}) T_g^4(t)}_{\text{ground}} \right. \\ & + \underbrace{F_{1-f}(\mathbf{x}) T_f^4(t)}_{\text{front building}} \\ & + \underbrace{F_{1-s}(\mathbf{x}) T_s^4(t)}_{\text{sky}} \\ & \left. + \underbrace{\int_{\Gamma_1} F_{1-1}(\mathbf{x}, \mathbf{x}') T^4(\mathbf{x}', t) d\Gamma_1(\mathbf{x}')}_{\text{wall itself}} \right), \end{aligned}$$

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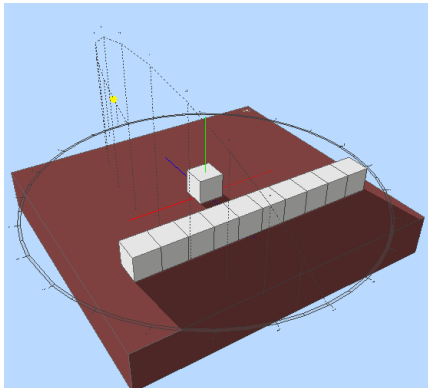


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Internal boundary conditions

Convection that varies with height, modeled using a corner model¹⁴:

$$h_{\infty,3}(y) = h_{30} + (h_{31} - h_{30}) \sin\left(\frac{\pi y}{H}\right),$$

Long-wave radiation is estimated by approximating the average radiant temperature of the environment with the indoor air temperature and the average temperature of the internal face of the wall:

$$q_{\infty,3}(\mathbf{x}, t) = \varepsilon_3 \sigma \left(T(\mathbf{x}, t)^4 - \left(\frac{1}{2} (T_{\infty,3}(t) + \langle T(\mathbf{x}, t) \rangle_{\Gamma_3}) \right)^4 \right).$$

$T_{\infty,3}$ is chosen as a function of the outdoor temperature $T_{\infty,1}$.

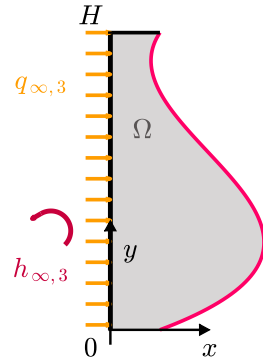


Figure 8: Internal boundary conditions.

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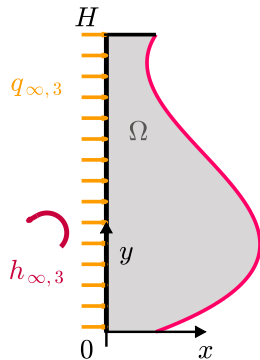


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Top and bottom boundary conditions

Adiabatic boundary conditions are applied at the top and bottom faces:

$$-\lambda \nabla T \cdot \mathbf{n} = 0, \quad \forall \mathbf{x} \in \Gamma_j, \quad j \in \{2, 4\}.$$

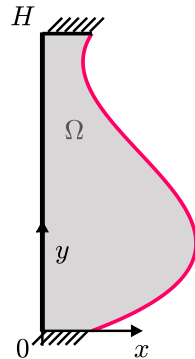


Figure 9: Internal boundary conditions.

Insulation domain optimization

Insulation domain definition

Interface Γ_i defined by implicit function γ :

$$\Gamma_i(\mathbf{p}) = \{ \mathbf{x} \in \Omega \mid \mathbf{x} = \gamma(s, \mathbf{p}), \mathbf{p} \in \Omega_p, s \in \Omega_s \}$$

with s the curvilinear abscissa and N_p parameters of the interface: $\mathbf{p} = (\mathbf{p}_1, \dots, \mathbf{p}_{N_p}) \in \Omega_p$.

$$\begin{aligned} \gamma : \Omega_s \times \Omega_p &\longrightarrow \mathbb{R}^2, \\ (s, \mathbf{p}) &\longmapsto (x = \gamma_x(s, \mathbf{p}), y = \gamma_y(s, \mathbf{p})), \end{aligned}$$

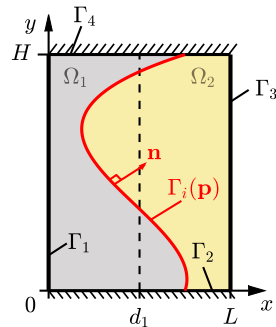


Figure 10: Illustration of the interface modelling with spline function.

Numerical method

- Finite difference method with uniform discretization in time and space ;
- DuFort-Frankel scheme known for its unconditional stability and accuracy¹⁵ ;
- Numerical integration at nodes located near the interface is based on the Ghost Fluid Method (GFM)¹⁶.

$$u_{j+}^n - u_{j-}^n = a_j^n,$$

$$\lambda_2 \frac{u_{j+1i}^n - u_{j+}^n}{(1-\theta)\Delta x} - \lambda_1 \frac{u_{j-}^n - u_{ji}^n}{\theta\Delta x} = b_j^n.$$

- Programming language: Matlab and Fortran ;

¹⁵Berger et al., "An efficient two-dimensional heat transfer model for building envelopes", 2020.

¹⁶Li et al., "Alternating direction ghost-fluid methods for solving the heat equation with interfaces", 2020.

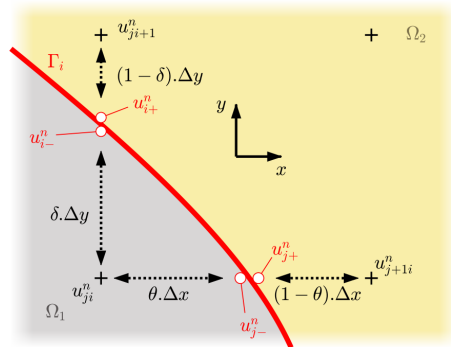


Figure 11: Illustration of simplified GFM.

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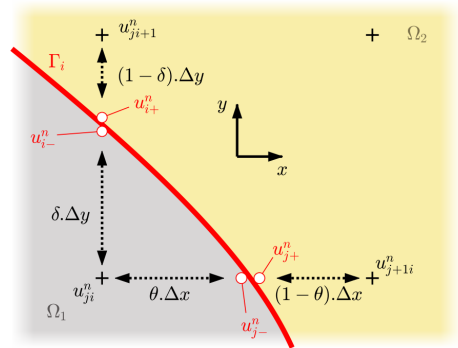


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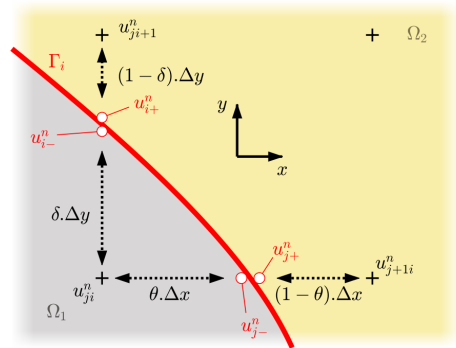


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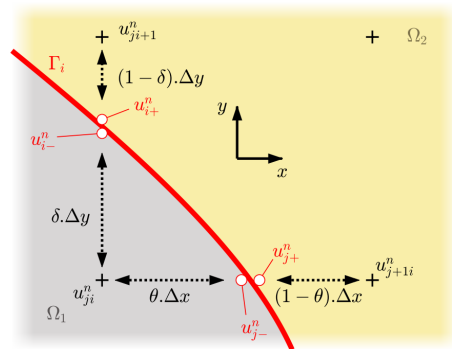


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Model accuracy

- Analytical solution:

$$u(\mathbf{x}, t) = \cos(2x) \sin(2y) \cos(t), \quad \forall \mathbf{x} \in \Omega_1$$

$$u(\mathbf{x}, t) = \sin(2x) \cos(2y) \cos(t), \quad \forall \mathbf{x} \in \Omega_2$$

with interface Γ_i where $\mathbf{p} = (0.05, 4)$:

$$\gamma(\mathbf{x}, \mathbf{p}) = 0.5 + p_0 \sin(p_1 \cdot \arctan(y, x)) - \sqrt{x^2 + y^2}$$

- At the interface between two materials: temperatures and conductive flux continuity depending on a jump condition.

$$u|_{\mathbf{x}+\epsilon} - u|_{\mathbf{x}-\epsilon} = a(\mathbf{x}, t), \quad \forall \mathbf{x} \in \Gamma_i$$

$$\lambda(\mathbf{x}) \frac{\partial u}{\partial \mathbf{n}^+} \Big|_{\mathbf{x}+\epsilon} - \lambda(\mathbf{x}) \frac{\partial u}{\partial \mathbf{n}^-} \Big|_{\mathbf{x}-\epsilon} = b(\mathbf{x}, t), \quad \forall \mathbf{x} \in \Gamma_i$$

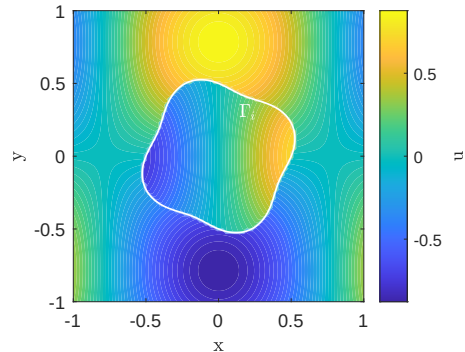


Figure 12: Solution of the model at $t = 0.5$.

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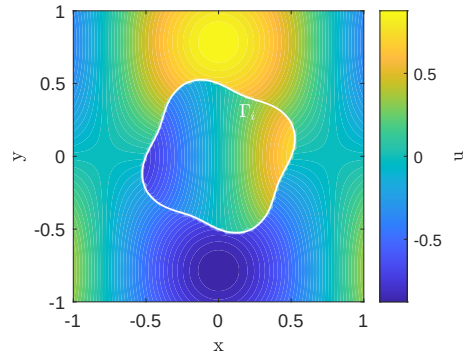


Figure 12: Solution of the model at $t = 0.5$.

Model accuracy

The accuracy of the model is assessed by computing both the L^2 and L^∞ norms of the error⁹.

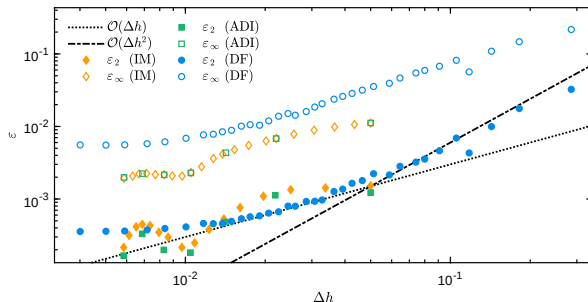


Figure 13: Variation of the numerical models errors according to the spatial steps ($\Delta h = \Delta x = \Delta y$) and compared to other models¹⁶.

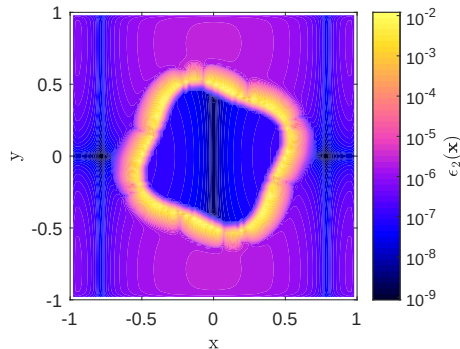


Figure 14: L^2 error field of the model.

⁹Gasparin et al., "An innovative method to determine optimum insulation thickness based on non-uniform adaptive moving grid", 2019.

Insulation domain definition

Use of BÉZIER parametric curves:

$$\gamma_x(s, \mathbf{p}) = \sum_{i=1}^{N_{\bullet}} \mathcal{B}_{i-1}^{N_{\bullet}}(s) \cdot x_i + A(\mathbf{p}),$$

$$\gamma_y(s, \mathbf{p}) = \sum_{i=1}^{N_{\bullet}} \mathcal{B}_{i-1}^{N_{\bullet}}(s) \cdot y_i, \quad \forall s \in [0, 1]$$

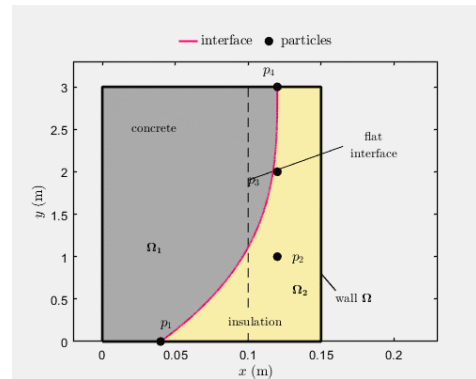


Figure 16: Illustration of the interface modelling with 4 particles BÉZIER curve.

Thermal design problem

Minimization problem: find the optimal coordinates of the control points: $p^* = \arg \min_{p \in \Omega_p} \mathcal{C}$ of the BÉZIER curve. $\mathcal{C}$ is based on the *net consumed work*⁵ w :

$$\mathcal{C} = \int_{t_i}^{t_f} w(t) dt, \quad \text{with} \quad w(t) = \left(\int_{\Gamma_3} -\lambda(\mathbf{x}) \cdot \frac{\partial T}{\partial x} dy \right) \left(\frac{T_{\infty,1}(t)}{T_{\infty,3}(t)} - 1 \right)$$

Constraints: conservation of the volumes of materials and interface contained in the wall.

Optimization strategy: NSGAI algorithm with convergence determined by relative objective variation and mean Euclidean parameter dispersion.

⁵Strub et al., "Second law analysis of periodic heat conduction through a wall", 2005.

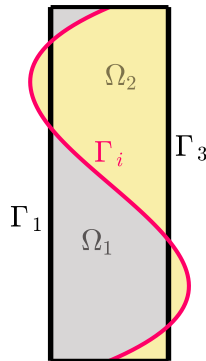
Thermal design problem

Minimization problem: find the optimal coordinates of the control points: $p^* = \arg \min_{p \in \Omega_p} \mathcal{C}$ of the BÉZIER curve. $\mathcal{C}$ is based on the *net consumed work*⁵ w :

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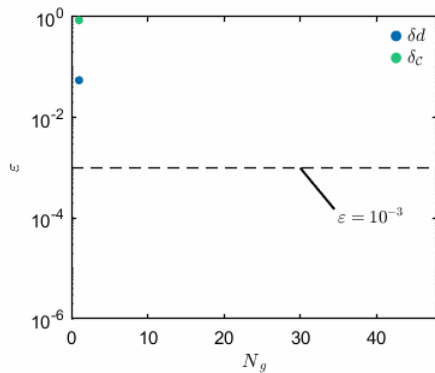
Constraints: conservation of the volumes of materials and interface contained in the wall.

Optimization strategy: NSGAI algorithm with convergence determined by relative objective variation and mean Euclidean parameter dispersion.

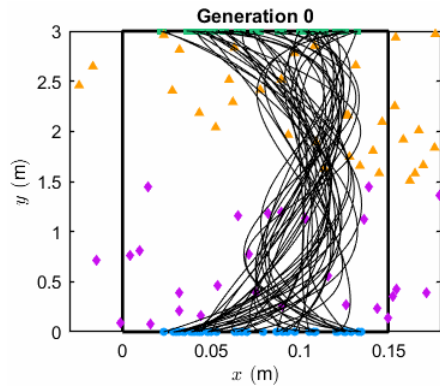


⁵Strub et al., "Second law analysis of periodic heat conduction through a wall", 2005.

Optimization results – Rio de Janeiro



(a)



(b)

Figure 17: Illustration of the convergence of the genetic algorithm, showing the convergence criteria (a) and the interfaces for each generation (b).

Optimization results – Rio de Janeiro

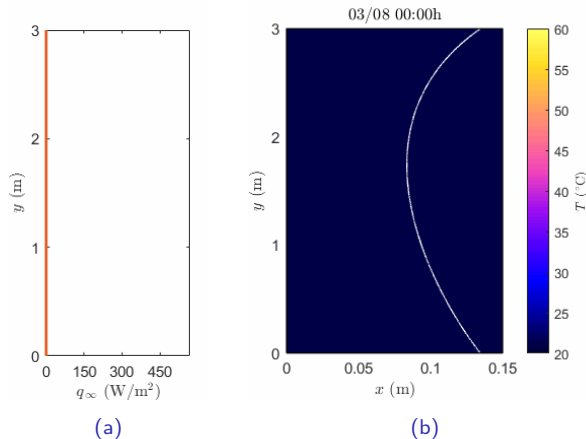
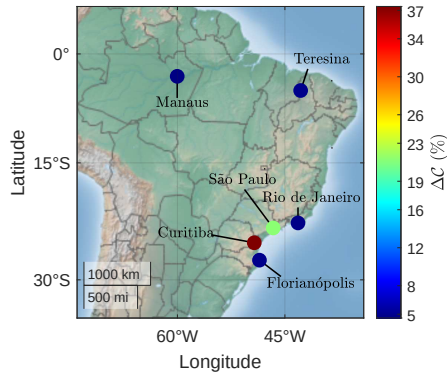
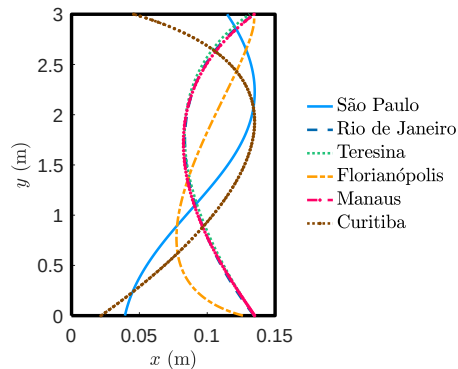


Figure 18: Illustration of the radiative external flux (a), and the temperature field in the wall (b) for the optimal configuration in Rio de Janeiro.

Optimization results – Brazil



(a)



(b)

Figure 19: Illustration of the relative energy improvement achieved after optimization (a), along with the corresponding optimal interface (b) for six Brazilian cities.

Shape optimization

Domain mapping

A coordinate transformation maps the physical domain Ω into a computational domain $\Omega^* = [0, 1]^2$ of coordinates $\mathbf{x}^* = (\xi, \eta)$:

$$\Psi^{-1} : \quad \Omega^* \longrightarrow \Omega$$

$$(\xi, \eta) \longmapsto (x = \xi \cdot \gamma(\eta, \mathbf{p}), y = \eta).$$

Here, γ is an explicit function, such as this spline:

$$\gamma(y, \mathbf{p}) = L + p_1 \frac{y}{H} \left(\frac{y}{H} - 1 \right) \left(\frac{y}{H} - p_2 \right) - \frac{p_1}{12} \left(2p_2 - 1 \right).$$

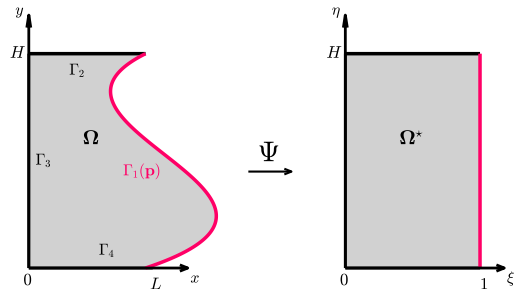


Figure 20: Illustration of physical and computational domains.

Domain mapping

2D thermal diffusion equation in the computational domain:

$$c \frac{\partial T}{\partial t} = \lambda \left(d_1(\mathbf{x}^*) \frac{\partial^2 T}{\partial \xi^2} + \frac{\partial^2 T}{\partial \eta^2} + d_2(\mathbf{x}^*) \frac{\partial^2 T}{\partial \xi \eta} + \underbrace{d_3(\mathbf{x}^*) \frac{\partial T}{\partial \xi}}_{\text{hyperbolic term}} \right), \quad \forall (\mathbf{x}^*, t) \in \Omega^* \times [0, t_f],$$

with:

$$d_1(\mathbf{x}^*) = \frac{1}{\gamma(\eta)^2} \left(1 + \xi^2 \gamma'(\eta)^2 \right),$$

$$d_2(\mathbf{x}^*) = \frac{-2\xi\gamma'(\eta)}{\gamma(\eta)},$$

$$d_3(\mathbf{x}^*) = \xi \left(\frac{2\gamma'(\eta)^2}{\gamma(\eta)^2} - \frac{\gamma''(\eta)}{\gamma(\eta)} \right).$$

Semi-discretization allows us to write the problem in the form of an ODE with the operator $\mathbf{M}$:

$$\frac{dT}{dt} = \mathbf{M}(T)$$

Numerical method

Runge–Kutta–Legendre or Runge–Kutta–Gegenbauer numerical schemes (RKL1, RKL2, RKG1 and RKG2):

$$U_0 = T(t^n)$$

$$U_1 = U_0 + \tilde{\mu}_1 \tau \mathbf{M} U_0$$

$$U_k = \mu_k U_{k-1} + \nu_k U_{k-2} + \gamma_k U_0 + \tilde{\mu}_k \tau \mathbf{M} U_{k-1} + \tilde{\gamma}_k \tau \mathbf{M} U_0, \quad 2 \leq k \leq s$$

$$T(t^n + \tau) = U_s,$$

with the parameters of the scheme μ_k , ν_k , $\tilde{\mu}_k$, $\tilde{\mu}_1$ and $\tilde{\gamma}_k$.

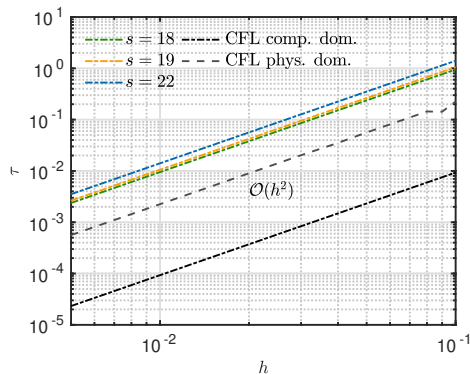
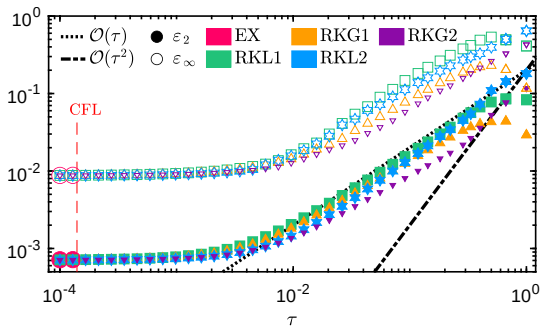
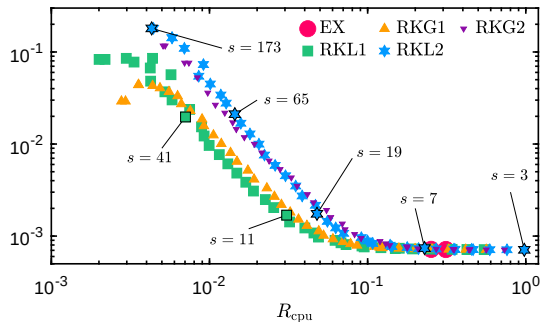


Figure 21: Illustration of the maximum time step to ensure stability of the RKL2 scheme according to the discretization mesh ($p_1 = 6.0$, $p_2 = 0.5$, $\alpha = 0.01$).

Accuracy of the model



(a)



(b)

Figure 22: Variation of the numerical models errors according to the time mesh (a) and the computational effort (b) for $h = 10^{-2}$. Results obtained for spline shape function ($p_1 = 6.0$, $p_2 = 0.5$) and an analytical solution¹⁷.

¹⁷Kadoch et al., "A volume penalization method for incompressible flows and scalar advection–diffusion with moving obstacles", 2012.

Thermal design problem

Minimization problem: find the optimal coordinates of the control points: $\mathbf{p}^* = \arg \min_{\mathbf{p} \in \Omega_p} \mathcal{C}$ of the spline curve. $\mathcal{C}$ is based on the:

$$\mathcal{C} = \int_{t_i}^{t_f} \left| \int_{\Gamma_3} -\lambda(\mathbf{x}) \cdot \frac{\partial T}{\partial x} dy \right|$$

Constraints: conservation of the volumes of the wall and the shape of Γ_1 does not overlap Γ_3 .

Optimization strategy: NSGAI algorithm with convergence determined by relative objective variation and mean Euclidean parameter dispersion.

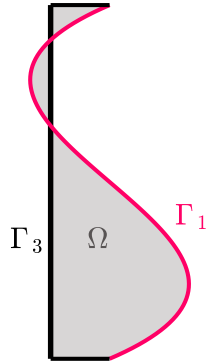
Thermal design problem

Minimization problem: find the optimal coordinates of the control points: $p^* = \arg \min_{p \in \Omega_p} \mathcal{C}$ of the spline curve. $\mathcal{C}$ is based on the:

$$\mathcal{C} = \int_{t_i}^{t_f} \left| \int_{\Gamma_3} -\lambda(\mathbf{x}) \cdot \frac{\partial T}{\partial x} dy \right|$$

Constraints: conservation of the volumes of the wall and the shape of Γ_1 does not overlap Γ_3 .

Optimization strategy: NSGAI algorithm with convergence determined by relative objective variation and mean Euclidean parameter dispersion.



Optimization results – France

Canyon street configuration in Lille with $\Delta X = 3$ m, $\Delta Y = 6$ m and a wall with height $H = 3$ m, thickness $L = 20$ cm. South-facing facade.

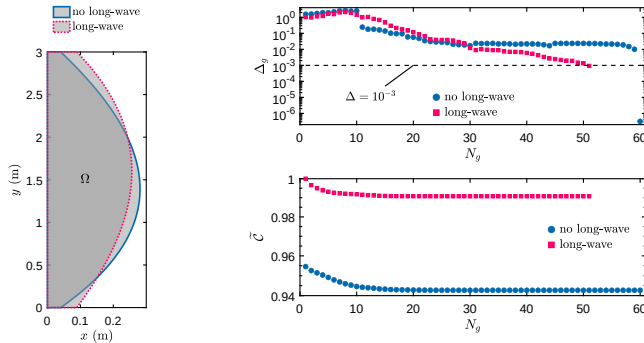


Figure 23: Optimal wall shape for modelling that does or does not take long-wave effects into account. Convergence of the criteria for the two optimisations and their functionals.

Optimization results – France

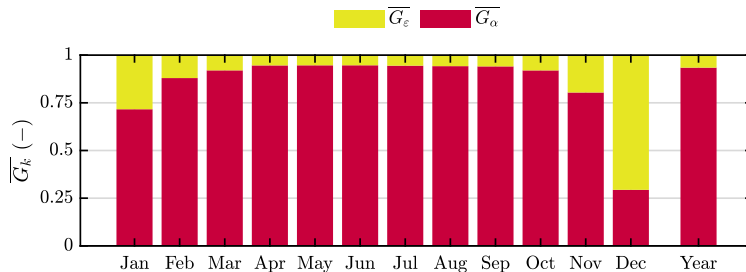


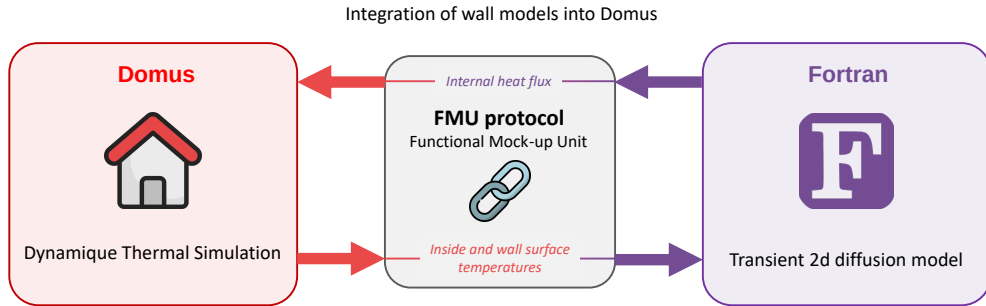
Figure 24: Global sensitivity¹⁸ of the absorptivity α and emissivity ϵ coefficients of the external surface.

¹⁸Sobol' and Kucherenko, "Derivative based global sensitivity measures and their link with global sensitivity indices", 2009.

Why do I come here ?

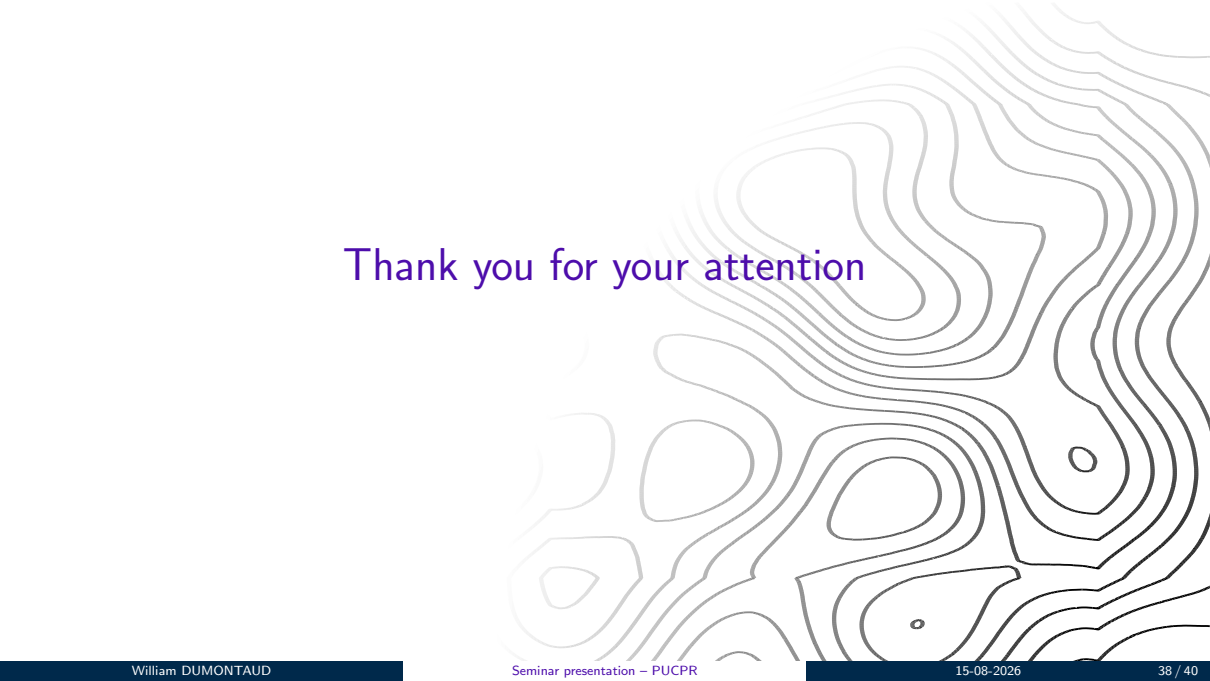
Current work objective

→ Scale up the models to the scale of the building simulation to make them more realistic:



→ Carry out an optimization (Kriging);

→ Develop a final model that enables the optimization of shape and topology.

A decorative background on the right side of the slide featuring a topographic map with various contour lines in shades of gray.

Thank you for your attention

Annexes

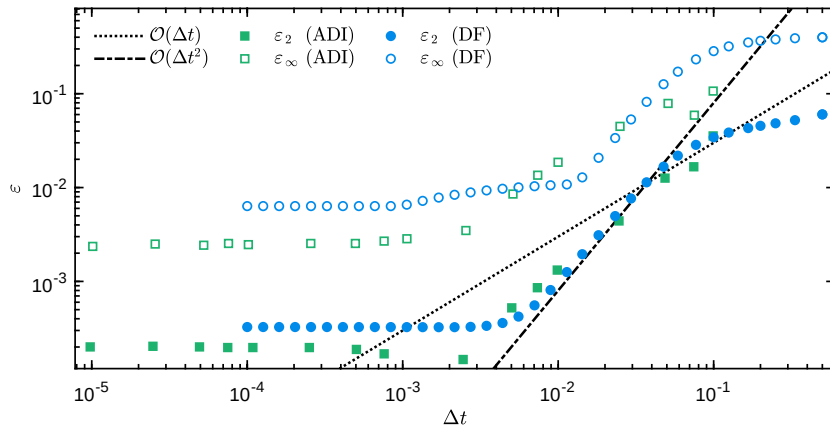


Figure 25: Variation of the numerical models errors according to the time discretisation.

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